



University of Vavuniya

Second Examination in Information Technology - 2022

First Semester - January/February 2024

IT2153 Computer Graphics (Practical)

Answer All Questions

Time Allowed: Two Hours

Instructions:

- Create a folder on the desktop with your index number and save all your files in it.
- You may see the output images in the given *Resources* folder stored in the shared folder in order to view the colour representations.
- It is your responsibility to save your programs frequently.

1. Write a Java program to perform each of the following tasks:

- (a) Implement the DDA line drawing algorithm to draw a straight line. [15%]
- (b) Construct the following initial object in red and blue using *MouseEvent* with the given specifications in pixels as shown in Figure 1 [Not to scale] using the line drawing algorithm in part (a) above:
[Note that the *MouseClicked* point indicated in Figure 1 below is the centre of a circle to draw for the question in part (e).]

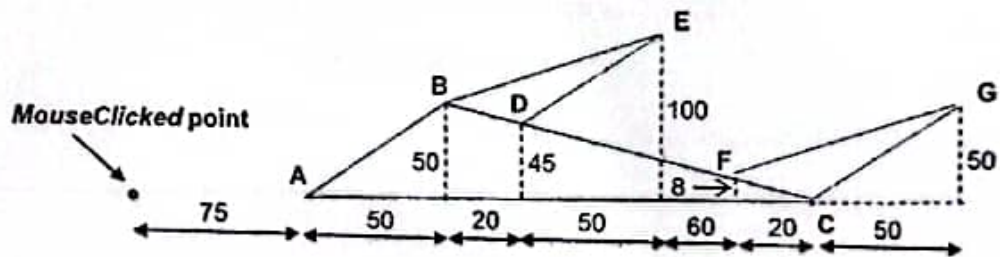


Figure 1

[20%]

- (c) Construct an object as shown in Figure 2 using the appropriate transformation(s) applied to the object constructed in part (b) above:

[15%]

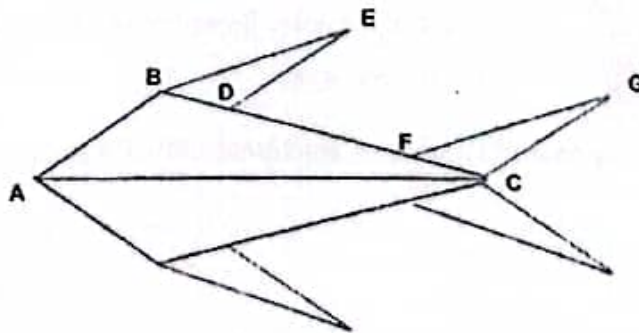


Figure 2

- (d) Implement *Bresenham's* circle drawing algorithm to draw a circle.

[20%]

[This question is continued on the next page.]

(e) Construct an object as shown in Figure 3 using appropriate transformation(s):

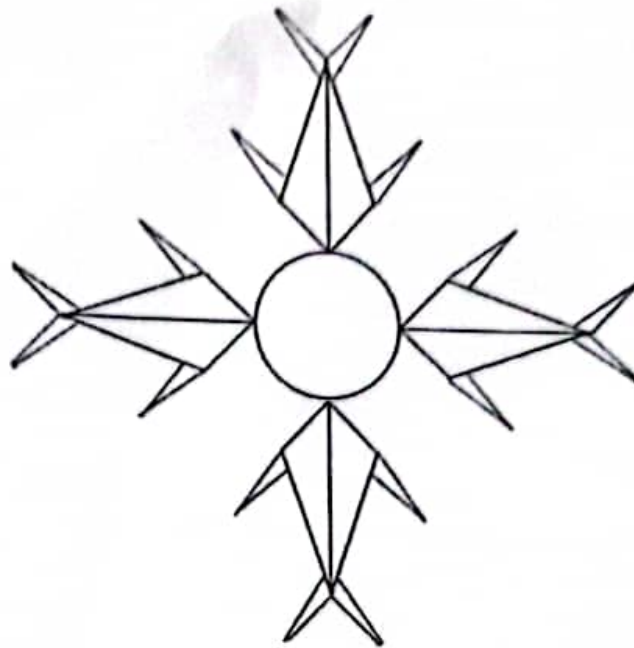


Figure 3

[30%]